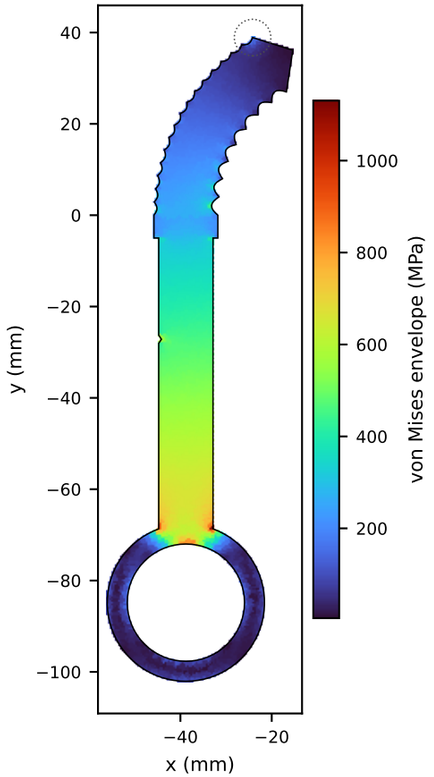
Wave 1 — smoke_constraints (first constrained evolve) | Fitness 12.256 N/g | Est. weight 34.6 g

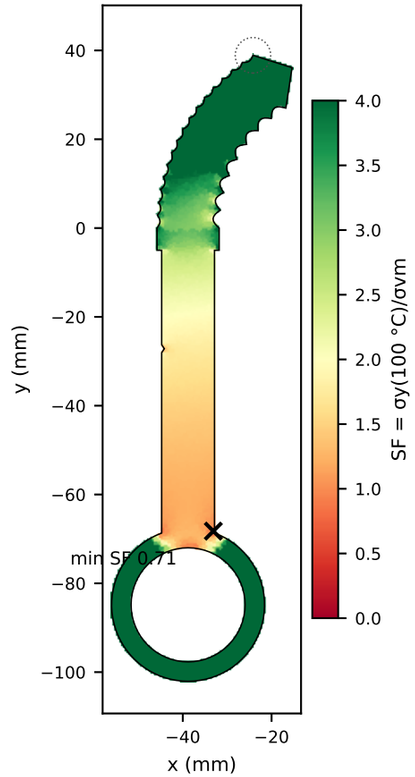
FEM-UNCALIBRATED: offline Kirchhoff plate (numerical DKT + CST, 4956 tris) — not yet calibrated to MC / repo FEM. Envelope of hunter-lateral 260 N, twist 1.26 N·m, tip-pry 260 N, ring-pull 1500 N.

FEM-UNCALIBRATED **QUARANTINED: blade < 3.375 in (all three)** **hard geometric gates: red-team Ok** **floors 180/260/1500 N on plate: FAIL**

(1) von Mises — worst load case per element



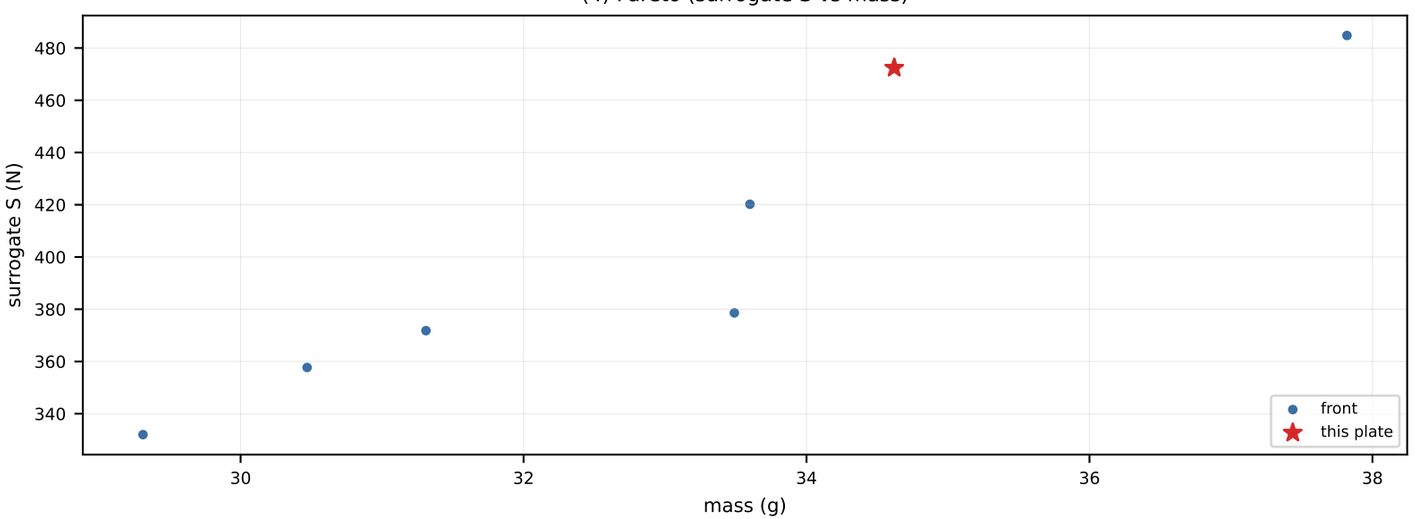
(2) Safety factor — worst case @100 °C



(3) Metrics

Metric	Value	Gate / note
Blade tip→heel	1.859 in (47.2 mm)	**≥3.375 in: FAIL**
Blade horizontal reach	1.197 in (30.4 mm)	**≥3.375 in: FAIL**
Blade edge arc (outer)	2.193 in (55.7 mm)	**≥3.375 in: FAIL**
Grip (ring-neck→guard)	63.7 mm	60–80 mm
Ring ID	25.68 mm	25.4±0.5 mm
Stock / material	4.4 mm / Ti-6Al-4V	red-team: Ok
Est. weight	34.62 g	geom check 34.62 g
Fitness	12.256 N/g	MC p05 strength-per-gram
Beam surrogate S	472 N	repo plate (fem.rs): n/a
Plate FEM @ -10 °C	lat 209 N | ring 5117 N	SF hiker 1.16 / hunter 0.80 / ring 3.41
Plate FEM @ +20 °C	lat 202 N | ring 4954 N	SF hiker 1.12 / hunter 0.78 / ring 3.30
Plate FEM @ +100 °C	lat 185 N | ring 4518 N	SF hiker 1.03 / hunter 0.71 / ring 3.01
Twist / pry SF @100 °C	twist 11.19 | pry 2.22	twist 1.26 N·m, pry 260 N

(4) Pareto (surrogate S vs mass)



Floors on offline plate, worst of -10/20/100 °C: hiker PASS · hunter FAIL · ring PASS
Temp: Ti yield linear derate (-0.11 %/°C from 20 °C); maps use 100 °C yield 803 MPa.
Serration knock-down on blade (h/h_eff=1.08, from history genes). Load patch r≤4 mm excluded from peaks.